Target Pressure and GDP Overreporting: Evidence from China's Poverty Alleviation Program

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Abstract

This paper investigates the effect of China's 2020 poverty alleviation goal on GDP reporting in poverty-stricken counties. We adapt Martínez's (2022) model to estimate differences in GDP-nighttime light elasticity between poverty and non-poverty counties, assuming this elasticity should be consistent across all counties without manipulation. Our findings reveal significant GDP overreporting in poverty-stricken counties following the announcement of the 2020 goal. We estimate these counties inflated their GDP figures by approximately 23%, meaning that for every 1% of actual economic growth, the officially reported GDP growth was about 1.23%. Event study analysis shows that overreporting increases as the target deadline approaches, suggesting escalating pressure on local officials to demonstrate performance. These results highlight the unintended consequences of top-down poverty alleviation targets and the potential for data manipulation in high-stakes policy environments.

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1 Introduction

The reliability of official statistics in policy-driven environments has long been a concern for both researchers and policymakers. This issue becomes particularly salient when ambitious policy targets create strong pressure on local officials to demonstrate performance. In November 2011, China launched one of history's most ambitious poverty alleviation campaigns by announcing a mandatory goal to eliminate absolute poverty by 2020. This campaign, affecting 832 poverty-stricken counties, provides an ideal setting to examine how target pressure might influence the quality of official economic statistics.

The Chinese context is particularly relevant for studying this issue for several reasons. First, the policy established a clear and rigid deadline (2020) for all poor counties to achieve a specific income threshold (2,300 RMB per capita in 2010 value), creating unprecedented pressure on local officials. Second, China's political system, where local officials' career advancement heavily depends on economic performance (Li and Zhou, 2005), amplifies the incentives for potential data manipulation. Third, while recent studies have documented the positive effects of this poverty alleviation campaign on rural income (Li et al., 2024), the potential unintended consequences on data quality remain unexplored.

This paper investigates whether and to what extent the pressure to meet poverty elimination targets leads to systematic overstatement of GDP growth in poverty-stricken counties. Following the methodological framework developed by Martínez (2022), we examine the relationship between nighttime lights and reported GDP across poor and non-poor counties before and after the 2012 policy announcement. Our key insight is that while both GDP and nighttime lights should reflect true economic activity, only GDP figures are subject to potential manipulation by local officials.

For our empirical analysis, we construct a comprehensive county-level panel dataset combining official statistics from China County Statistical Yearbooks with satellite-based nighttime light data. We utilize a harmonized global nighttime light dataset developed by Li

et al. (2020), which provides consistent measurements from 1992 to 2018 by integrating observations from both the Defense Meteorological Satellite Program (DMSP) and the Visible Infrared Imaging Radiometer Suite (VIIRS). This harmonized dataset allows us to track changes in economic activity through an objective lens that is immune to local manipulation.

Our findings reveal significant evidence of GDP overreporting in poverty-stricken counties following the announcement of the 2020 poverty elimination goal. Using our preferred specification with full controls, we estimate that these counties inflated their GDP figures by approximately 38.3% after the TPA policy implementation. In other words, for every 1% of actual economic growth (as proxied by changes in nighttime lights), the officially reported GDP growth was about 1.383%. Through an event study analysis, we further document that this overreporting intensifies as the 2020 deadline approaches, with the elasticity difference between poor and non-poor counties growing steadily from 2012 and reaching approximately 0.03 by 2020. This temporal pattern suggests escalating pressure on local officials to demonstrate progress toward the poverty elimination target. The gradual increase in manipulation magnitude not only provides strong evidence for our hypothesis but also helps rule out alternative explanations, as any confounding factors would need to perfectly match this dynamic pattern of intensifying effects as the deadline approached.

The manipulation effects are particularly pronounced in counties with weaker fiscal conditions. Counties with above-median government expenditure-to-GDP ratios or deficit-to-GDP ratios show substantially larger degrees of GDP overreporting, suggesting that financial constraints might amplify the incentives for data manipulation. These heterogeneous effects provide additional evidence supporting our interpretation that the observed patterns indeed reflect strategic responses to policy pressure rather than other confounding factors.

This paper makes several contributions to the literature. Firstly, we contribute to the growing literature on the political economy of data manipulation (Chen et al., 2016; Martínez, 2022) by providing the first systematic evidence of how poverty alleviation targets affect the quality of local GDP statistics in China. While previous studies have documented

various forms of statistical manipulation in response to political incentives, our work uniquely identifies how specific policy deadlines can intensify such behavior.

Secondly, we contribute to the broader literature on the effectiveness of target-based governance (Li and Zhou, 2005; Xu, 2011) by demonstrating its potential unintended consequences. Our findings suggest that while setting explicit targets may motivate local officials, it can also compromise the quality of statistical reporting, particularly as deadlines approach. This insight has important implications for the design of performance evaluation systems in hierarchical bureaucracies.

Finally, our results speak to the growing literature on China's poverty alleviation efforts (Li et al., 2024) by highlighting an overlooked aspect of the campaign's impact. While previous research has focused primarily on the program's effectiveness in reducing poverty, our study reveals the complex interplay between policy targets, political incentives, and statistical integrity.

2 Policy Background

China's poverty alleviation policies can be traced back to 1986 when the government established nationally designated poor counties as the main targets of poverty alleviation efforts, initially identifying 331 counties. In 1994, with the introduction of the "8-7 National Poverty Reduction Program," the number of poor counties was finalized at 592, following a comprehensive review based on the "Four in, Seven out" principle. This program aimed to ensure that by 2000, rural households in poverty would achieve an annual per capita net income of 500 RMB, focusing on improving infrastructure such as drinking water, electricity, and healthcare in rural areas. In November 2011, China's central government announced an ambitious goal of eliminating absolute poverty by 2020. This announcement set a clear deadline for local governments and established a specific poverty threshold—an annual per capita income of 2,300 RMB (2010 value) in rural areas. Following this announcement, 832 counties were identified as poor counties for poverty alleviation. Among these, in March 2012,

the State Council Leading Group Office of Poverty Alleviation and Development designated 592 counties as national key poor counties to receive more concentrated assistance and resources.

The program's implementation framework involves multiple levels of government coordination. The central government provides policy guidance and financial support through fiscal transfer payments. Provincial governments supervise the implementation process and assess local officials' performance in poverty reduction. At the county level, local governments of all 832 poor counties are tasked with implementing various poverty alleviation measures and demonstrating tangible progress toward the 2020 target.

Implementing this poverty elimination target created strong pressure on local officials, particularly in poor counties. Unlike previous poverty alleviation programs in China that mainly focused on general economic development (Park and Wang, 2010), this new campaign established clear accountability measures for local governments. Between 2012 and 2015, the performance evaluation for poverty alleviation efforts in poor counties primarily used GDP as a key indicator. However, after 2016, the evaluation criteria shifted to focus on improving the living standards of the poor population and reducing the number of people in poverty, with the removal of GDP assessment for ecologically fragile and protected areas. The State Council Leading Group Office of Poverty Alleviation and Development initiated an exit mechanism for poor counties to address cases where counties with substantial economic improvements were reluctant to leave the list. The exit standards included per capita disposable income, public recognition, and poverty incidence rates, with a multi-level review and approval process spanning county, municipal, provincial, and central levels. However, remaining on the list of poor counties allowed continued access to financial subsidies, incentivizing local officials to manipulate economic data by adjusting the extent of misreporting or delaying declarations of poverty alleviation progress to prolong access to subsidies.

This strict target-based system, with a clear deadline and a hierarchical accountability structure, creates substantial pressure on local officials in poor counties. The pressure is

particularly intense because the target is absolute (complete poverty elimination) rather than relative improvement, and the deadline is rigid (by 2020) with no flexibility. Li and Zhou (2005) have shown that local officials' career prospects in China are closely tied to economic performance indicators. The institutional setting of China's political system adds another layer of complexity to this issue. As Chen et al. (2019) demonstrate in their forensic examination of China's national accounts, the combination of high-powered incentives and limited external oversight can create conditions conducive to data manipulation.

Recent research has documented various positive outcomes of the Targeted Poverty Alleviation (TPA) program. Li et al. (2024) found that the program increased rural income in national key poor counties by approximately 10.9%. Zhang, Xie and Zheng (2023) demonstrate that the program's relocation policy significantly improved household income and children's educational outcomes. Research by Luo et al. (2022) has also confirmed the program's effectiveness in increasing residents' income. However, Huang and Zhu (2021) point out that while this multi-level implementation framework effectively mobilized resources, it also generated specific policy implementation issues, particularly regarding imbalances in the incentive mechanisms during local government policy execution.

Of particular note is the complex dual-incentive structure facing local officials under this policy framework: on the one hand, poverty alleviation performance was directly linked to officials' performance evaluation and career advancement, motivating them to promote poverty reduction actively; on the other hand, maintaining poor county status enabled continued access to substantial resources such as financial subsidies. This conflicting incentive mechanism could lead local officials to engage in strategic behavior in data reporting and poverty alleviation progress declarations.

Our research examines whether the implementation of this poverty elimination target leads to systematic overstatement of GDP growth in poor counties. Our main analysis focuses on all 832 poor counties identified for poverty alleviation. As a robustness check, we also examine the subset of 592 national key poor counties that received more intensive policy

treatment. The selection of poor counties was based on several criteria, including baseline income levels, fiscal conditions, geographic features (such as location in mountainous areas or old revolutionary bases), and the proportion of rural population. By comparing the relationship between nighttime lights and reported GDP growth across poor and non-poor counties before and after the 2011 policy announcement, we aim to detect and quantify potential GDP manipulation driven by poverty elimination pressure.

3 Data and Sample

4 Data

Our analysis combines three main data sources: satellite-based nighttime light data, county-level economic statistics, and county characteristics from various administrative sources. The temporal coverage spans from 2000 to 2020, providing sufficient observations both before and after the 2011 poverty alleviation policy announcement.

Our primary proxy for actual economic activity comes from satellite-based nighttime light observations. We utilize a harmonized dataset that integrates observations from two satellite systems: the Defense Meteorological Satellite Program's Operational Linescan System (DMSP-OLS) and the Visible Infrared Imaging Radiometer Suite (VIIRS). Following the methodology developed by Li et al. (2020), we perform sensor calibration and conversion of DMSP-OLS data collected before 2012 to ensure compatibility with NPP-VIIRS observations collected after April 2013. This integration process generates an extended NPP-VIIRS dataset that maintains consistency and accuracy across the entire study period. The resulting nighttime light measures provide a consistent and manipulation-free indicator of economic activity at the county level. We aggregate the pixel-level light intensity to county boundaries using GIS techniques, creating annual measures of total light output for each county.

Our official economic statistics come primarily from the China County Statistical Year-

books (2000-2020). These yearbooks provide detailed information on various economic indicators, with our main variable of interest being county-level GDP. To address variations in statistical reporting across different years and regions, we supplement the yearbook data with information from the China County Statistical Database and official statistical bureaus of local governments, which provide additional verification and missing data points. Our dataset includes comprehensive economic indicators such as nominal and real GDP, government fiscal revenue and expenditure, population and demographic indicators, as well as other economic measures including industrial output and fixed asset investment. To ensure comparability across time, all monetary values are adjusted to 2010 constant prices using provincial-level GDP deflators.

4.1 Sample

Our sample consists of all county-level administrative units in mainland China for which we have consistent data coverage. The treatment group comprises 832 counties officially designated as poverty-stricken areas under the poverty alleviation campaign. For additional analysis, we also identify a subset of 592 national key poor counties that received more intensive policy intervention.

In constructing our final sample, we implement several selection criteria to ensure data quality and consistency. We exclude counties that experienced administrative boundary changes during our sample period to ensure consistent geographical units for analysis. Furthermore, we remove observations with missing values for key variables including GDP, nighttime lights, and fiscal conditions. To mitigate the influence of outliers, we also exclude counties with extreme values in either GDP growth or nighttime light growth, defined as those falling in the top and bottom one percentile of the distribution.

For our main analysis, we use all non-poor counties as the control group. However, recognizing that poor and non-poor counties may differ systematically, we also construct a matched control group based on pre-treatment characteristics. The matching process consid-

ers initial economic conditions including GDP per capita and industrial structure, geographic features such as terrain type and distance to major cities, demographic characteristics including population density and rural population share, and fiscal conditions measured by the government revenue-to-GDP ratio. This matching approach helps ensure that our treatment and control groups are comparable along observable dimensions that might affect both economic performance and reporting incentives.

5 Empirical Specification

In this section, We present the econometric model that guides the empirical analysis. Following the methodological framework developed by Martínez (2022) for detecting GDP manipulation in autocracies, We adapt the model to examine potential GDP growth overstatement in China's poverty-stricken counties under the Targeted Poverty Alleviation (TPA) policy. Martínez's approach exploits the heterogeneity in the relationship between nighttime lights (NTL) and GDP across political regimes to detect manipulation. We extend this strategy by examining the heterogeneity in NTL-GDP elasticity between poverty and non-poverty counties before and after the TPA policy implementation, which set a mandatory poverty elimination target by 2020.

Let the unobserved variable $\tilde{y}_{i,t}$ represent true economic growth in county i during year t. Following Martínez (2022), We decompose growth into components, but instead of comparing democracies and autocracies, We allow for the possibility that true growth differs between non-poverty and poverty-stricken counties. This involves a baseline growth rate for non-poverty counties ($y_{n,i,t}$) and an adjustment factor α for poverty-stricken counties ($p_{i,t} = 1$):

$$\tilde{y}_{i,t} = y^n_{i,t} + \alpha p_{i,t}. \tag{1}$$

The local government of each county produces an estimate of economic growth using GDP. We assume that the estimated GDP growth rate, $g_{i,t}$, is a linear function of true

income growth and an error term $\epsilon_{i,t}$. However, this estimate may not match the reported GDP growth rate, $\hat{g}_{i,t}$, which could be subject to manipulation in poverty-stricken counties after the implementation of TPA policy in 2012. In the following equation, We allow for the possibility that poverty-stricken counties overstate GDP growth by an amount proportional to the true observed growth rate after TPA implementation ($TPA_t = 1$):

$$g_{i,t} = \beta \tilde{y}_{i,t} + \epsilon_{i,t}. \tag{2}$$

$$\hat{g}_{i,t} = (1 + \sigma p_{i,t} \times TPA_t) g_{i,t}. \tag{3}$$

Following the literature on using nighttime lights as a proxy for economic activity (Henderson, Storeygard and Weil, 2012), We assume that the growth rate of NTL ($l_{i,t}$) is also a linear function of true income growth and a separate error term $u_{i,t}$. Importantly, the satellite-based NTL data is independently collected and processed, making it immune to manipulation by local governments.

$$l_{i,t} = \gamma y^n_{i,t} + \gamma \alpha p_{i,t} + u_{i,t}. \tag{4}$$

Combining these equations yields:

$$\hat{g}_{i,t} = \frac{\beta}{\gamma} l_{i,t} + \frac{\beta\sigma}{\gamma}(l_{i,t} \times p_{i,t} \times TPA_t) + \left(\lambda + \sigma\epsilon_{i,t} - \frac{\sigma\beta}{\gamma} u_{i,t}\right) p_{i,t} + \sigma\lambda(p_{i,t} \times TPA_t) + \nu_{i,t}, \tag{5}$$

where $\lambda \equiv (1 - \frac{\gamma_p}{\gamma_n})\beta\alpha$ and ν is a combination of the error terms ϵ and u.

Following Henderson, Storeygard and Weil (2012), We rewrite the equation in log-linear form and specify the main estimating equation as:

$$\ln(\text{GDP})_{i,t} = \mu_i + \delta_t + \phi_1 \ln(\text{Light})_{i,t} + \phi_2(\ln(\text{Light})_{i,t} \times p_{i,t} \times \text{TPA}_t) + \mathbf{X}_{i,t}\beta + \xi_{i,t}, \tag{6}$$

where μ_i is a county fixed effect, δ_t is a year fixed effect, and $\xi_{i,t}$ is the error term clustered by county. The vector $\mathbf{X}_{i,t}$ includes time-varying county characteristics. The coefficient ϕ_2 captures the differential effect of NTL on GDP in poverty-stricken counties after the TPA policy implementation, with a positive estimate suggesting potential GDP growth overstatement.

Similar to Martínez's approach of using regime type variation, We exploit the variation in poverty status and policy timing to identify potential manipulation. While Martínez (2022) focused on cross-country variation in political institutions, We focus on within-country variation in local government incentives created by the TPA policy's strict poverty elimination targets.

To interpret $\phi_2 > 0$ as evidence of GDP manipulation, We follow Martínez (2022) in assuming that the NTL elasticity of GDP does not systematically differ across units in the absence of manipulation. However, while Martínez needed to address cross-country heterogeneity, We focus on potential systematic differences between poverty and non-poverty counties within China that might affect the mapping of NTL growth to GDP growth. We address this through various robustness checks, including controlling for county characteristics and using alternative definitions of poverty status. Additionally, extending Martínez's approach of examining variation in manipulation incentives, We provide evidence that the elasticity difference is larger when the pressure to meet poverty alleviation targets is stronger, such as near the 2020 deadline.

6 Results

6.1 Baseline Estimates

Table 1 shows the main results of the paper. The dependent variable in all columns is $ln(GDP)$, and all regressions include county and year fixed effects. Column 1 reports an average night light elasticity of GDP equal to 0.107, which is consistent with previous

literature. To detect potential GDP manipulation in poverty-stricken counties after the implementation of the Targeted Poverty Alleviation (TPA) policy in 2012, We gradually introduce additional controls in columns 2-5.

The coefficient of interest is the interaction between $ln(Light)$ and the TPA policy indicator. Following the model presented in Section 4, this coefficient captures the differential effect of night lights on GDP in poverty-stricken counties after the TPA policy implementation. A positive value for this coefficient suggests that a same-sized increase in night lights is associated with a larger increase in reported GDP in poverty counties under the TPA policy, consistent with GDP manipulation. The ratio $\frac{\phi_2}{\phi_1}$ provides a measure of GDP growth overstatement, where ϕ_2 is the coefficient on the interaction term and ϕ_1 is the baseline elasticity. For instance, in my preferred specification with full controls (column 5), $\frac{0.018}{0.047} \approx 0.383$ suggests that poverty counties overstate GDP growth by approximately 38.3% after the TPA policy implementation.

The results are remarkably stable across specifications. Column 2 includes provincial industrial structure controls, column 3 adds other provincial characteristics, column 4 introduces county-level industrial structure controls, and column 5 includes county-level characteristics. The interaction coefficient remains positive and statistically significant at the 1% level across all specifications, ranging from 0.012 to 0.018. This indicates that under the TPA policy, a 1% increase in night lights is associated with a reported GDP growth that is 1.2-1.8 percentage points higher in poverty counties compared to non-poverty counties.

6.2 Heterogeneous Effects

Table 2 further explores the heterogeneity in GDP manipulation through subgroup analysis. We focus on whether the government's relative influence over local economic activity affects their ability to manipulate GDP statistics. Specifically, we split the sample based on fiscal indicators measured in 2007, our sample's initial year.

Panels A and B compare counties with high versus low government expenditure. Using

the median ratio of government expenditure to GDP in 2007 as the cutoff, we find significant GDP manipulation only in counties with above-median expenditure ratios. This pattern is consistent with our expectation that counties where government spending constitutes a larger share of GDP have greater capacity to influence reported economic statistics, as a substantial portion of GDP components are directly under government control. In the full specification (column 5), the interaction coefficient is 0.014 and significant at the 1% level for the high-expenditure group, implying approximately $\frac{0.014}{0.042} \approx 33.3\%$ overstatement of GDP growth. In contrast, the interaction coefficient is close to zero and insignificant for the low-expenditure group, where private sector activity plays a relatively larger role in the local economy.

Panels C and D present results by government deficit levels. Similarly divided by the median deficit-to-GDP ratio in 2007, counties with higher deficits also show evidence of GDP manipulation under the TPA policy. This finding further supports our hypothesis about the role of government influence, as higher deficit-to-GDP ratios typically indicate more active government intervention in the local economy through deficit spending. The interaction coefficient in the full specification is 0.014 and significant at the 1% level for the high-deficit group ($\frac{0.014}{0.039} \approx 35.9\%$ overstatement), while it remains close to zero and insignificant for the low-deficit group.

These heterogeneous effects reveal an important pattern in how local governments respond to policy pressure through statistical manipulation. Counties where government expenditure or deficit accounts for a larger share of GDP show substantially higher levels of GDP overreporting (33.3-35.9%), while counties with smaller government footprints show no significant evidence of manipulation. This pattern suggests that the ability to manipulate GDP statistics is closely tied to the government's relative weight in the local economy. When government activities constitute a larger portion of local GDP, officials have greater capacity to influence reported economic figures, likely because they have more direct control over the underlying components of GDP calculation. This finding highlights the crucial role of economic structure in determining local governments' ability to respond to policy target

pressure through statistical manipulation.

6.3 Event Study Design

To estimate the dynamic effects of the TPA policy on GDP manipulation, We modify the baseline specification to allow for year-by-year changes in the night light elasticity of GDP. Given that China's central government announced the 2020 poverty elimination target in November 2011, We use 2010 as the reference year to avoid any anticipation effects. Specifically, We estimate the following equation:

$$\begin{aligned} ln(GDP)_{i,t} =& \mu_i + \delta_t + \phi_1 ln(Light)_{i,t} + \sum_{k=-5}^{8} \beta_k [ln(Light)_{i,t} \times Poor_i \times 1(t - 2012 = k)] \\ &+ \mathbf{X}_{i,t}\gamma + \xi_{i,t} \end{aligned} \tag{7}$$

where μ_i is the county fixed effect, δ_t is the year fixed effect, $Poor_i$ is an indicator for poverty-stricken counties, and $1(t-2012 = k)$ are indicator variables for each year k relative to the TPA implementation in 2012. The vector $\mathbf{X}_{i,t}$ includes the same set of control variables as in the baseline specification, including provincial and county-level industrial structure and other socioeconomic characteristics. The coefficients of interest are the β_k's, which capture the differential night light elasticity of GDP between poverty and non-poverty counties in each year relative to 2012.

We normalize β_{-2} to zero, making 2010 the reference year. This choice is particularly important given the timing of the poverty elimination announcement in late 2011, which could have already started influencing local officials' reporting behavior. Therefore, each β_k represents the change in the night light elasticity of GDP for poverty counties relative to non-poverty counties, compared to their difference in 2010. The pattern of β_k before 2010 provides a test of the parallel trends assumption, while the pattern after 2012 reveals the dynamic effects of the TPA policy.

Figure 1 plots these β_k coefficients and their 95% confidence intervals. The estimation framework proves particularly informative in several dimensions. The flat and insignificant pre-2010 coefficients provide strong support for the parallel trends assumption, suggesting that poverty and non-poverty counties exhibited similar reporting patterns before any policy intervention. We observe a slight uptick in the coefficient during 2011, consistent with potential anticipation effects following the central government's announcement. The coefficients then show a clear and steady increase after the formal implementation of TPA in 2012, reaching approximately 0.03 by 2020. This pattern demonstrates the growing extent of GDP manipulation in poverty counties as they approached the poverty elimination deadline.

7 Conclusion

This paper examines how China's 2020 poverty alleviation target affected the quality of local economic statistics. Using satellite-based nighttime light data as an objective measure of economic activity, we investigate potential GDP manipulation in poverty-stricken counties following implementing the Targeted Poverty Alleviation (TPA) policy. Our analysis combines county-level panel data from China County Statistical Yearbooks (2000-2020) with harmonized nighttime light observations from multiple satellite systems, allowing us to track changes in economic activity through a lens immune to local manipulation.

Our findings reveal significant evidence of GDP overreporting in poverty-stricken counties after the announcement of the 2020 poverty elimination goal. Using our preferred specification with full controls, we estimate that these counties inflated their GDP figures by approximately 38.3% following the implementation of the TPA policy. The manipulation effects are particularly pronounced in counties with weaker fiscal conditions, where government expenditure or deficit accounts for a larger share of GDP. Through an event study analysis, we document that this overreporting intensifies as the 2020 deadline approaches, with the elasticity difference between poor and non-poor counties growing steadily from 2012 and reaching approximately 0.03 by 2020.

These results highlight the unintended consequences of setting rigid policy targets in hierarchical bureaucracies. While China's poverty alleviation campaign has achieved remarkable success in reducing rural poverty, our findings suggest that the pressure to meet these targets has led to systematic distortions in official statistics, potentially compromising the ability to assess policy outcomes and allocate resources efficiently and accurately.

Looking forward, our findings provide important implications for policy formulation and implementation. While the target responsibility system has played a crucial role in mobilizing and incentivizing local governments, overly strict target assessments may lead to data distortion, affecting the accuracy of policy effectiveness evaluation. Future research could further explore optimizing the target assessment mechanism by incorporating diversified evaluation indicators and improved monitoring systems while maintaining policy incentives to enhance statistical data reliability. These insights are essential for China's poverty reduction policies and provide valuable references for poverty alleviation practices in other developing countries.

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Tables

Table 1: The Night Light Elasticity of GDP and the Targeted Poverty Alleviation Policy

	(1)	(2)	(3)	(4)	(5)
$ln(Light)$	0.107***	0.103***	0.063***	0.058***	0.047***
	(0.020)	(0.019)	(0.012)	(0.018)	(0.013)
$ln(Light) \times TPA$	0.018***	0.016***	0.017***	0.012***	0.018***
	(0.005)	(0.005)	(0.004)	(0.004)	(0.003)
Year fixed effects	√	√	√	√	√
County fixed effects	√	√	√	√	√
Provincial industrial structure		√	√	√	√
Provincial characteristics			√	√	√
County industrial structure				√	√
County characteristics					√

Note:

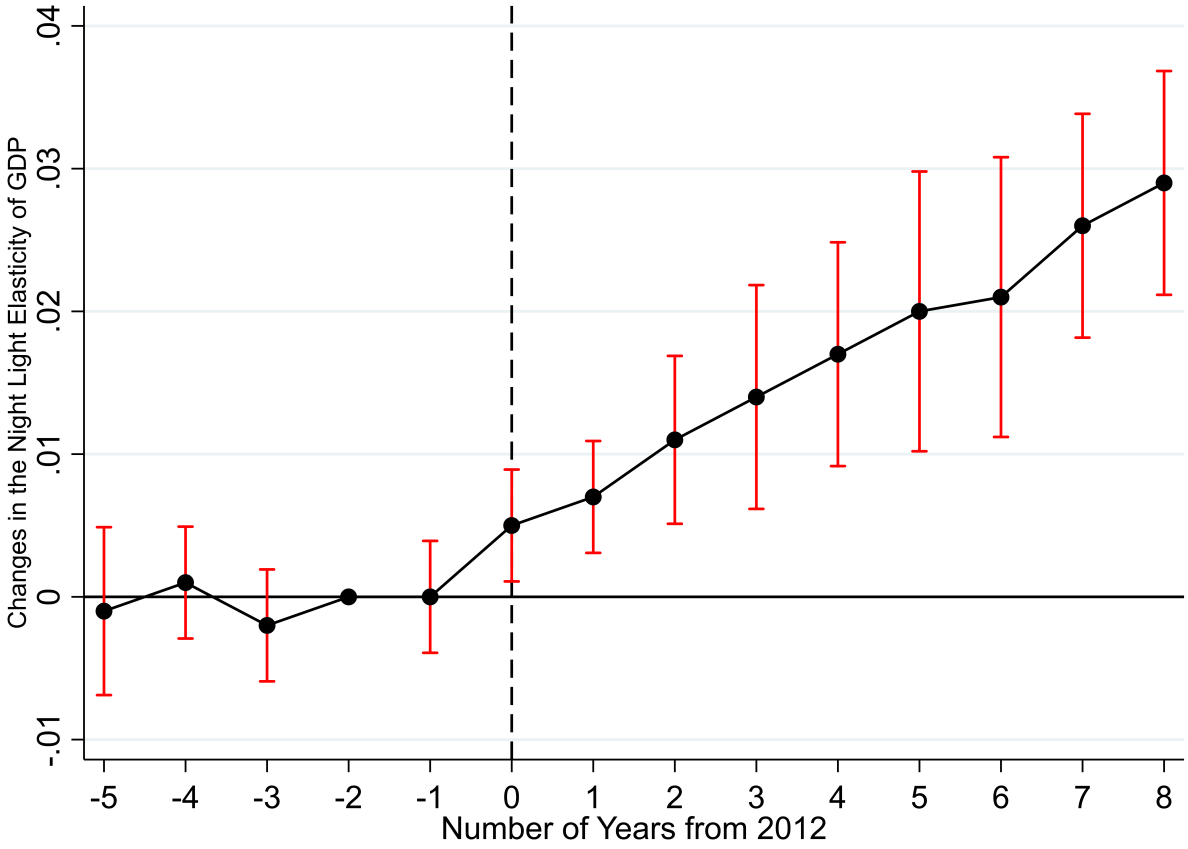
Table 2: The Night Light Elasticity of GDP and the Targeted Poverty Alleviation Policy: Subgroup Analysis

	(1)	(2)	(3)	(4)	(5)
Panel A: *High Government Expenditure*					
$ln(Light)$	0.089***	0.088***	0.049***	0.055**	0.042**
	(0.023)	(0.022)	(0.013)	(0.024)	(0.016)
$ln(Light) \times TPA$	0.013***	0.011**	0.010**	0.010***	0.014***
	(0.004)	(0.004)	(0.004)	(0.003)	(0.002)
# of Observation			10,752		
Panel B: *Low Government Expenditure*					
$ln(Light)$	0.148***	0.138***	0.088***	0.074***	0.062***
	(0.019)	(0.018)	(0.012)	(0.012)	(0.009)
$ln(Light) \times TPA$	-0.004	-0.002	-0.000	-0.004	0.000
	(0.004)	(0.004)	(0.003)	(0.004)	(0.003)
# of Observation			10,752		
Panel C: *High Deficit*					
$ln(Light)$	0.085***	0.084***	0.047***	0.051**	0.039**
	(0.022)	(0.021)	(0.013)	(0.022)	(0.015)
$ln(Light) \times TPA$	0.012**	0.010*	0.009**	0.011**	0.014***
	(0.005)	(0.005)	(0.004)	(0.004)	(0.003)
# of Observation			10,752		
Panel D: *Low Deficit*					
$ln(Light)$	0.149***	0.138***	0.086***	0.075***	0.066***
	(0.020)	(0.018)	(0.012)	(0.014)	(0.011)
$ln(Light) \times TPA$	-0.006	-0.004	-0.004	0.000	0.001
	(0.004)	(0.004)	(0.003)	(0.004)	(0.003)
# of Observation			10,752		
Year fixed effects	√	√	√	√	√
County fixed effects	√	√	√	√	√
Provincial industrial structure		√	√	√	√
Provincial characteristics			√	√	√
County industrial structure				√	√
County characteristics					√

Note:

Figure

Figure 1: The Night Light Elasticity of GDP



Note: